

KNEES AND THIGHS

Most of the muscles that control the movements of the knee are found in the thigh. A few calf muscles, however, are also involved. Generally, the thigh muscles that move the knee are categorized into two groups. Four large anterior thigh muscles (rectus femoris, vastus intermedius, vastus lateralis, vastus medialis) are collectively called the quadriceps muscles; these are the major knee extensors. Three large posterior thigh muscles (biceps femoris, semimembranosus, semitendinosus) are collectively called the hamstrings; these are the major knee flexors. The hamstrings are assisted in knee flexion by other thigh muscles (gracilis and sartorius) and a few calf muscles (gastrocnemius, popliteus, plantaris). Figures showing these muscles as well as a chart showing specific movements for each muscle are located at the end of the chapter (pages 110-111).

The muscles of the thigh that control the knee are important in all motor movements. Being much larger than the muscles of the calf and the foot, the thigh muscles are better able to withstand muscular stress. Hence, muscular soreness occurs less often in these muscle groups. It is important, however, to have the right balance of strength and flexibility between the opposing muscle groups of the thigh. Most people have stronger but less flexible quadriceps muscles than hamstring muscles. People tend to stretch the hamstring muscles much more than the quadriceps muscles. This creates an imbalance between the two muscle groups. Chronic overstretching of the hamstrings without comparable stretching of the quadriceps can cause more harm than good. This is the reason hamstring muscles are sore more often than quadriceps muscles. Overstretching can also lead to chronic fatigue and a decrease in strength in the hamstring muscles. To correct this imbalance, you need to put more emphasis on quadriceps stretching and decrease the emphasis on hamstring stretching.

People often sit in one particular position for a long time (as when riding in a car, sitting behind the desk at work, or sitting in an airplane seat). Thus, it is not surprising that people, after sitting for hours, feel the need to get up and stretch their muscles. When people do stand up after long periods of sitting, they typically find that their joints and muscles have become temporarily stiff. Stretching these muscles is a natural remedy. Many have found that stretching provides relief from muscular tension and pain. Since muscular soreness and tension are common in the thigh muscles, both temporary and lasting relief can be obtained from a regular daily stretching routine. This routine needs to be a consistent part of a fitness program.

Most of the instructions and illustrations in this chapter are given for the right side of the body. Similar but opposite procedures would be used for the left side of the body.

Standing Knee Flexor Stretch



Technique

Stand upright with the right heel 1 to 2 feet (30 to 61 cm) ahead of the left toes.

Keeping the right knee straight and the left knee slightly bent, bend the trunk over toward the right knee.

Reach the hands toward the right foot.

Muscles Stretched

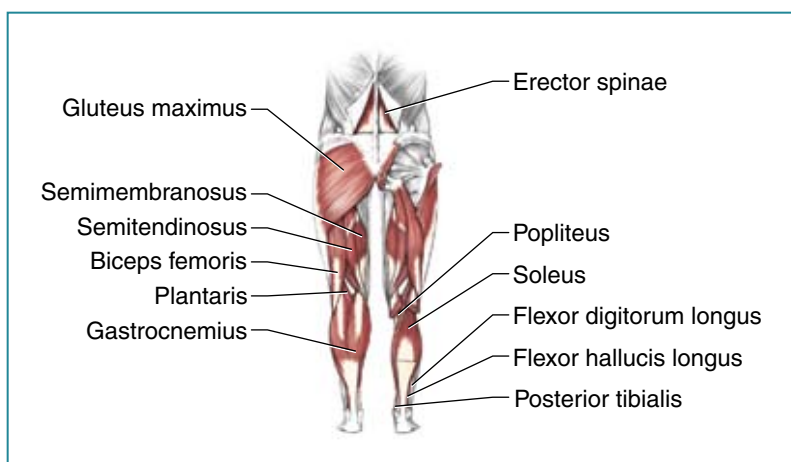
Most-stretched muscles: Right semitendinosus, right semimembranosus, right biceps femoris, right gluteus maximus, right gastrocnemius, lower erector spinae.

Lesser-stretched muscles: Right soleus, right plantaris, right popliteus, right flexor digitorum longus, right flexor hallucis longus, right posterior tibialis.

Commentary

For the best stretch, keep the right knee straight and bend the trunk directly from the hip. Keep the back as straight as possible. Turning the right foot slightly outward and bending the head and trunk more toward the medial (inner) side of the right knee will increase the stretch of the biceps femoris. Turning the right foot slightly inward and bending the head and trunk more toward the lateral (outer) side of the knee will increase the stretch of the semitendinosus and semimembranosus muscles.

Seated Knee Flexor Stretch



Technique

Sit on the floor with the legs extended and the inside of the ankles as close together as possible.

Keep the feet relaxed in a natural position.

Place the hands on the floor next to the thighs.

Bend at the waist and lower the head toward the legs. If possible, keep the back of the knees on the floor.

While bending forward, slide the hands toward the feet and keep them alongside the legs.

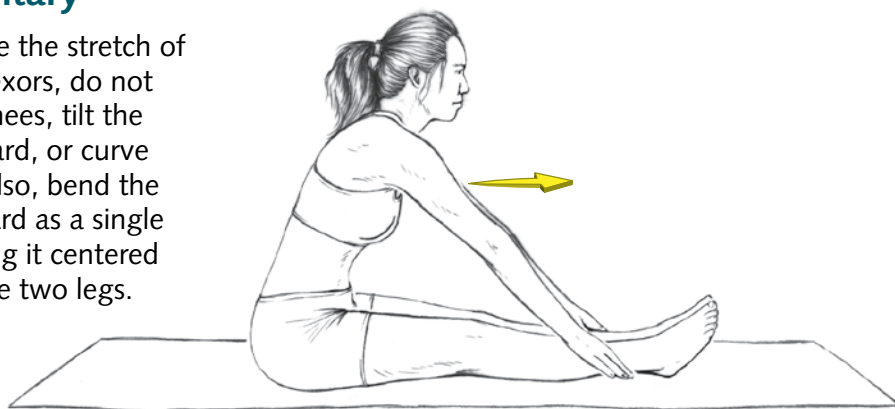
Muscles Stretched

Most-stretched muscles: Semitendinosus, semimembranosus, biceps femoris, gluteus maximus, gastrocnemius, lower erector spinae.

Lesser-stretched muscles: Soleus, plantaris, popliteus, flexor digitorum longus, flexor hallucis longus, posterior tibialis.

Commentary

To maximize the stretch of the knee flexors, do not bend the knees, tilt the pelvis forward, or curve the back. Also, bend the trunk forward as a single unit, keeping it centered between the two legs.



VARIATION

Seated Knee, Ankle, Shoulder, and Back Stretch

Changing the hand position to grasp the toes shifts the stretch to other muscles.

Technique

Sit on the floor with the legs extended and the inside of the ankles as close together as possible.

Keep the feet relaxed in a natural position.

Bend at the waist and lower the head toward the legs. If possible, keep the back of the knees on the floor.

While bending forward, slide the hands toward the feet, grasp the feet, and pull the toes slowly toward the knees (dorsiflexed position).

Muscles Stretched

Most-stretched muscles: Semitendinosus, semimembranosus, biceps femoris, gluteus maximus, gastrocnemius, lower erector spinae, soleus, plantaris, popliteus, flexor digitorum longus, flexor hallucis longus, posterior tibialis.

Lesser-stretched muscles:

Lower latissimus dorsi, lower trapezius, posterior deltoid, teres major, teres minor, infraspinatus, triceps brachii.



Raised-Leg Knee Flexor Stretch

Technique

Stand upright with weight balanced on the left leg.

Flex the right hip and place the right leg (with the knee straight) on a table, bench, or other stable object that is approximately the same height as the hips.

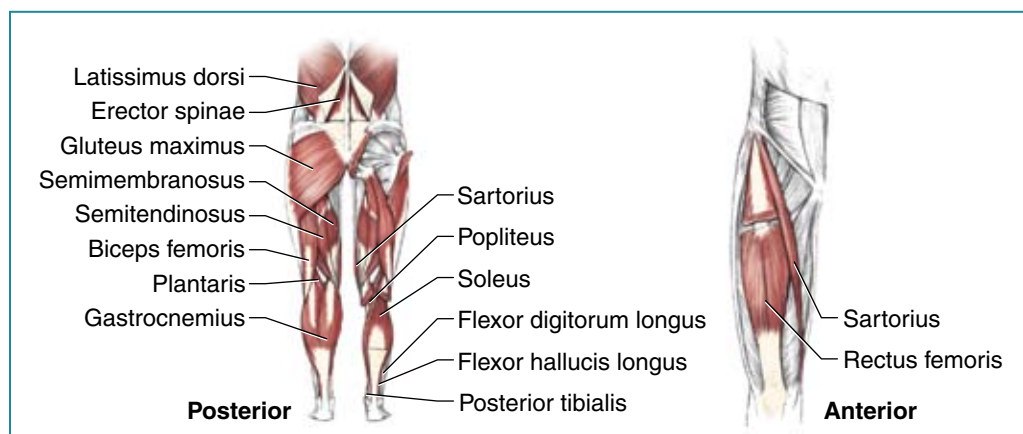
Bend at the waist, extend your arms over the lower right leg, and lower the head toward the right leg, keeping the right knee as straight as possible.

Keep the left knee straight and the left foot pointing in the same direction as the right leg.



Muscles Stretched

Most-stretched muscles: Right gluteus maximus, right semitendinosus, right semimembranosus, right biceps femoris, erector spinae, lower latissimus dorsi, right gastrocnemius.



Lesser-stretched muscles: Right soleus, right popliteus, right plantaris, right flexor digitorum longus, right flexor hallucis longus, right posterior tibialis, left sartorius, left rectus femoris.

Commentary

To maximize the stretch of the knee flexors, do not bend the knees, tilt the pelvis forward, or curve the back. Also, bend the trunk straight forward as a single unit, keeping it centered over the right leg.

Increasing the height of the table or bench by 1 to 2 feet (30 to 61 cm) above the hips will increase the stretch of these muscle groups. You will also start feeling a stretch in some of the left-side muscle groups (sartorius, rectus femoris, vastus intermedius, lateralis, and medialis) as you increase the height of the table.

VARIATION

Raised-Leg Knee, Ankle, Shoulder, and Back Stretch

Changing the hand position to grasp the toes shifts the stretch to other muscles.

Technique

Stand upright with weight balanced on the left leg.

Flex the right hip and place the right leg (with the knee straight) on a table, bench, or other stable object that is approximately the same height as the hips.

Bend at the waist, extend your arms over the lower right leg, and lower the head toward the right leg, keeping the right knee as straight as possible.

While bending forward, slide the hands toward the feet, grasp the feet, and pull the toes slowly toward the knees (dorsiflexed position).

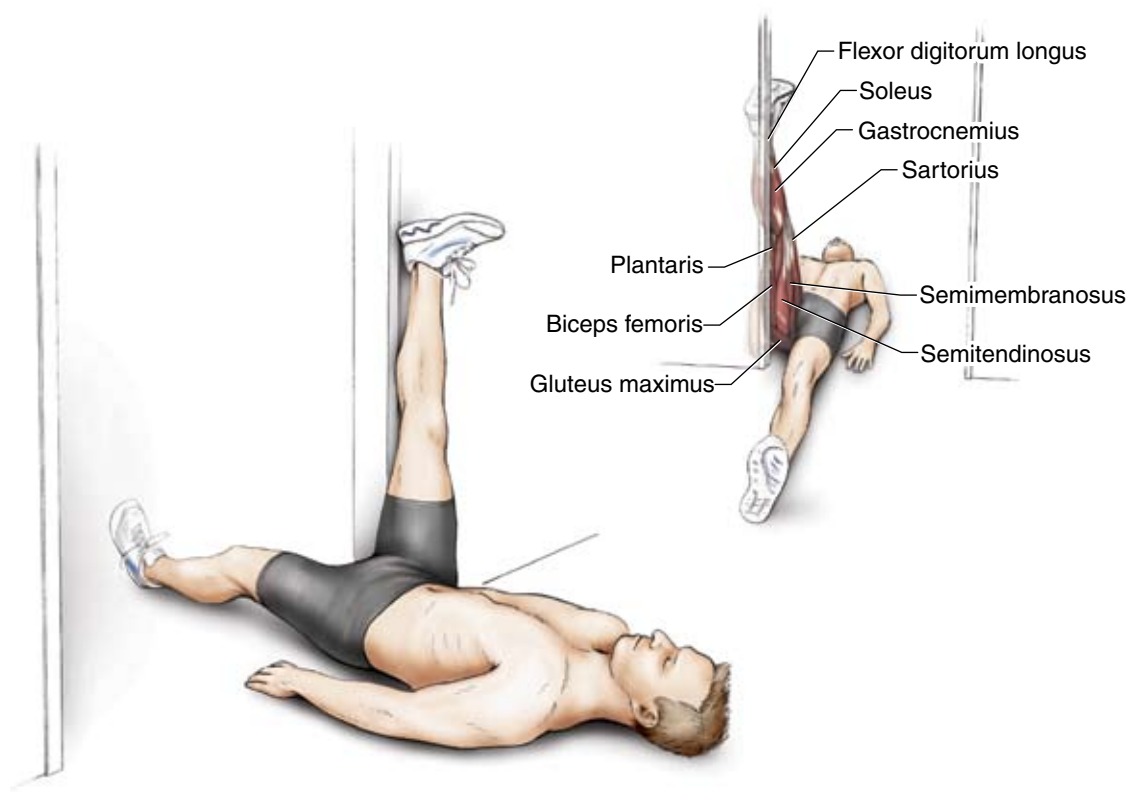


Muscles Stretched

Most-stretched muscles: Right gluteus maximus, right semitendinosus, right semimembranosus, right biceps femoris, erector spinae, lower latissimus dorsi, right gastrocnemius, right soleus, right popliteus, right plantaris, right flexor digitorum longus, right flexor hallucis longus, right posterior tibialis.

Lesser-stretched muscles: Left sartorius, left rectus femoris, lower trapezius, posterior deltoid, teres major, teres minor, infraspinatus, triceps brachii.

Recumbent Knee Flexor Stretch



Technique

Lie flat on your back in a doorway with the hips placed in front of the doorframe.

Raise the right leg and rest it on the doorframe. Keep the right knee straight and the left leg flat on the floor.

Place the hands palms down on either side of the buttocks.

Keeping the right leg straight, use the hands to slowly move the buttocks through the doorframe until you feel a stretch in the back of the leg.

Muscles Stretched

Most-stretched muscles: Right gluteus maximus, right semitendinosus, right semimembranosus, right biceps femoris, right gastrocnemius.

Lesser-stretched muscles: Right soleus, right popliteus, right plantaris, right flexor digitorum longus, right flexor hallucis longus, right posterior tibialis, left sartorius, left rectus femoris.

Commentary

To maximize the stretch of the knee flexors, do not bend the knees, tilt the pelvis forward, or curve the back. Adjust the distance between the buttocks and the doorframe to increase or decrease the stretch. The closer the buttocks are to the doorframe, the greater the stretch. Once the buttocks cannot be positioned any closer to the doorframe, bending the leg at the hip and moving the leg toward the head can increase the stretch.

VARIATION

Recumbent Knee, Ankle, Shoulder, and Back Stretch

Using a towel to bend the toes shifts the stretch to other muscles.

Technique

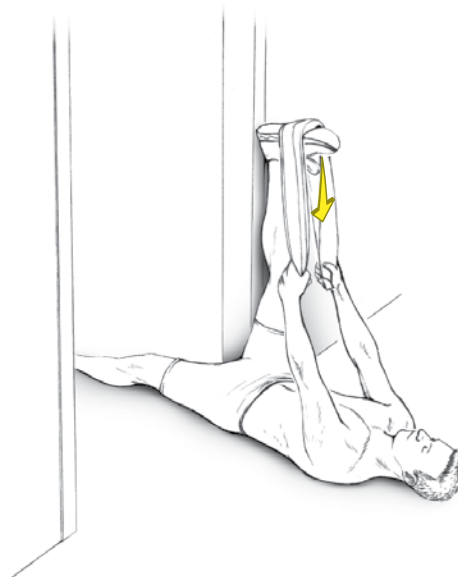
Lie flat on your back in a doorway with the hips placed in front of the doorframe.

Raise the right leg and rest it on the doorframe. Keep the right knee straight and the left leg flat on the floor.

Place a towel, cloth, or band over and around the toes and grasp both ends firmly with the hands.

Keeping the right leg straight, place the hands on the doorframe and slowly move the buttocks through the doorframe until you feel a stretch in the back of the leg.

Once you feel the stretch in the back of the leg, use the towel to pull the toes and foot down toward the head.

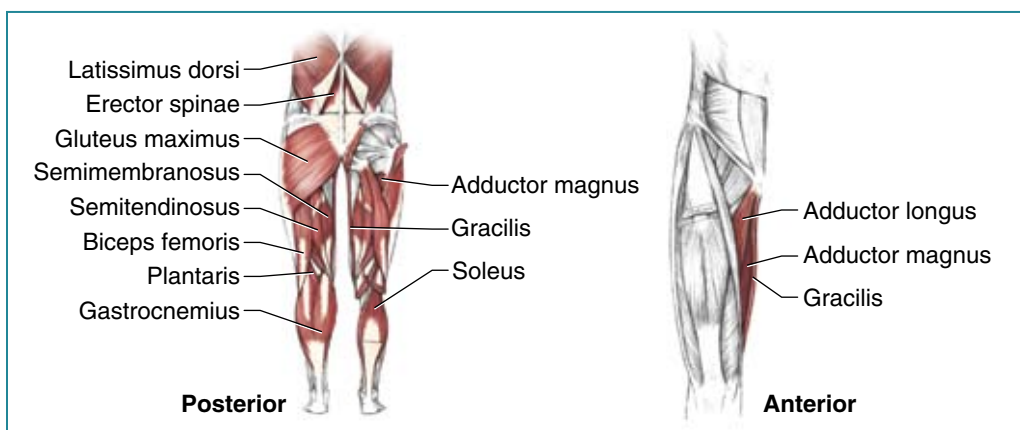
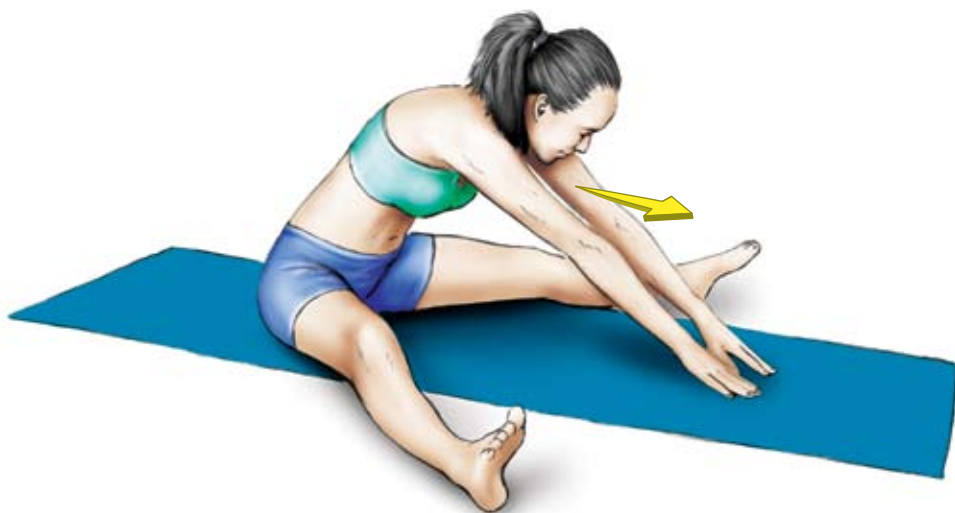


Muscles Stretched

Most-stretched muscles: Right gluteus maximus, right semitendinosus, right semimembranosus, right biceps femoris, erector spinae, lower latissimus dorsi, right gastrocnemius, right soleus, right popliteus, right plantaris, right flexor digitorum longus, right flexor hallucis longus, right posterior tibialis.

Lesser-stretched muscles: Left sartorius, left rectus femoris, lower trapezius, posterior deltoid, teres major, teres minor, infraspinatus, triceps brachii.

Seated Knee Flexor and Hip Adductor Stretch



Technique

Sit comfortably on the floor with legs extended in a V position (feet far apart from each other).

Place the hands on the floor next to the thighs.

Keep both knees straight and as flat against the floor as possible. Reach the hands out toward the center or slide the hands forward along the legs and bend the trunk over between the knees.

Muscles Stretched

Most-stretched muscles: Semitendinosus, semimembranosus, gracilis, adductor magnus and longus, gluteus maximus, lower erector spinae, lower latissimus dorsi, medial side of soleus, medial head of gastrocnemius.

Lesser-stretched muscles: Lateral soleus, lateral head of gastrocnemius, plantaris, biceps femoris.

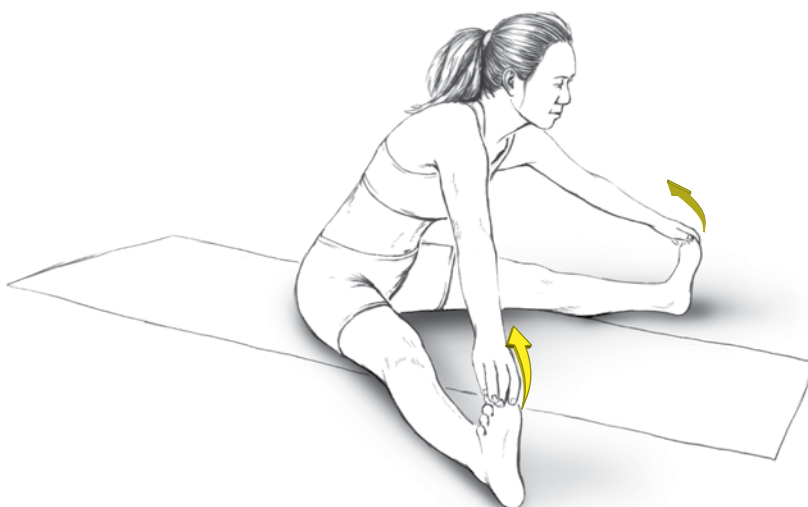
Commentary

To maximize the stretch, do not bend the knees, tilt the pelvis forward, or curve the back. Also, bend the trunk forward as a single unit, keeping it centered between the two legs.

VARIATION

Seated Knee, Hip, Ankle, Shoulder, and Back Stretch

Changing the hand position to grasp the toes shifts the stretch to other muscles.



Technique

Sit comfortably on the floor with legs extended in a V position (feet far apart from each other).

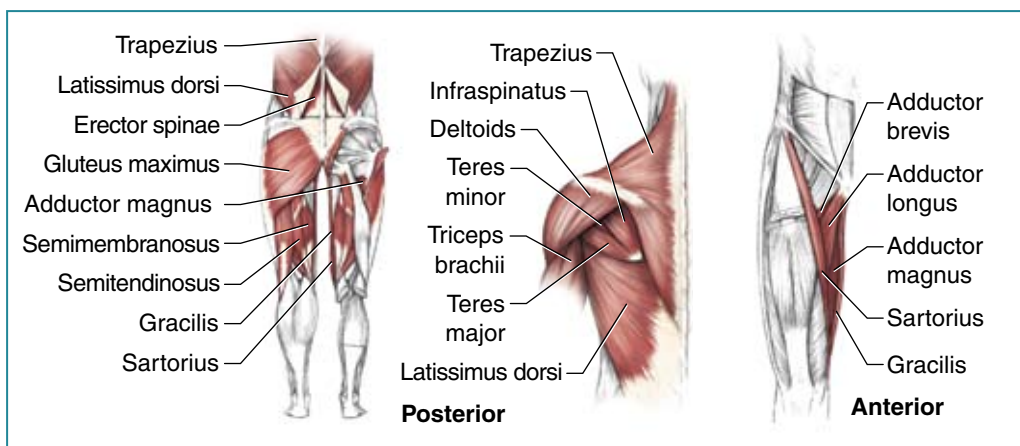
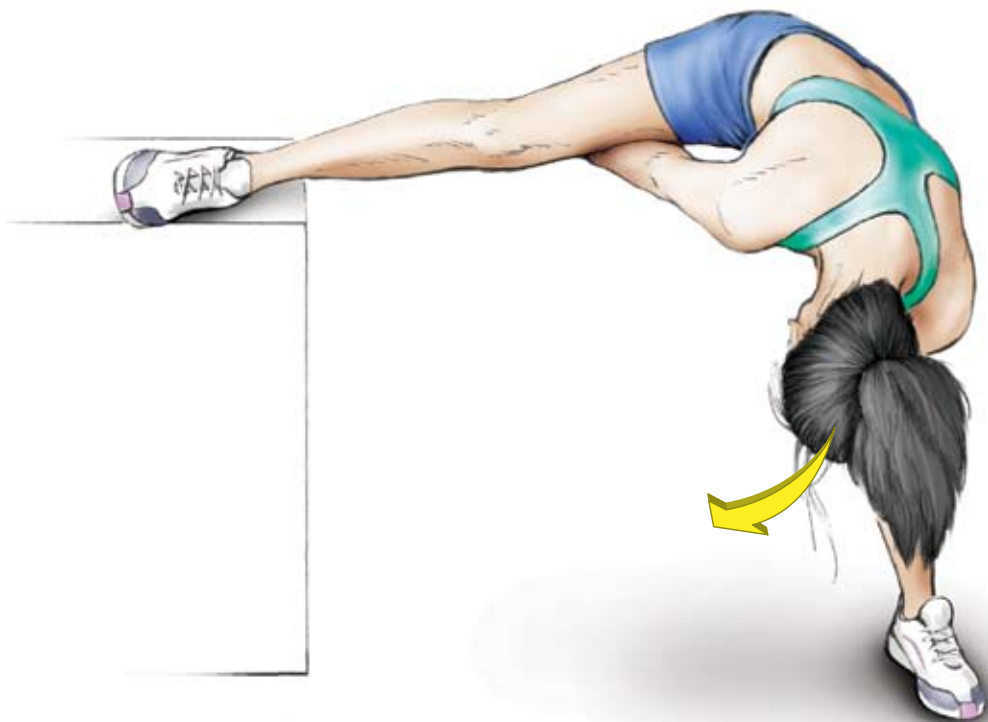
Keep both knees straight and as flat against the floor as possible. Slide the hands forward along the legs and bend the trunk over between the knees. At the same time, grasp the toes of both feet and pull them toward the body.

Muscles Stretched

Most-stretched muscles: Semitendinosus, semimembranosus, gracilis, adductor magnus, adductor longus, gluteus maximus, lower erector spinae, lower latissimus dorsi, soleus, gastrocnemius, popliteus, plantaris, flexor digitorum longus, flexor hallucis longus, posterior tibialis.

Lesser-stretched muscles: Biceps femoris, posterior deltoid, triceps brachii, teres major, teres minor, infraspinatus, lower trapezius.

Standing Knee Flexor and Hip Adductor Stretch



Technique

Stand with the right side of the body near a table, bench, or object at the approximate height of the hips.

While balancing the weight on the left leg, raise the right leg and position it on the table, bench, or object.

Keeping the knees straight, bend the trunk over between the knees as far as possible.

As you bend the trunk, reach between the knees and place the hands behind the thighs.

Muscles Stretched

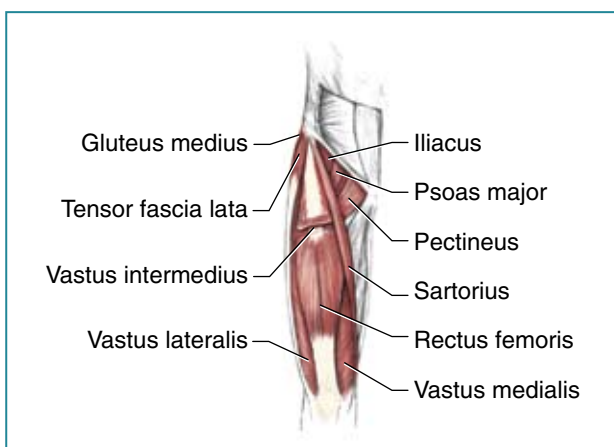
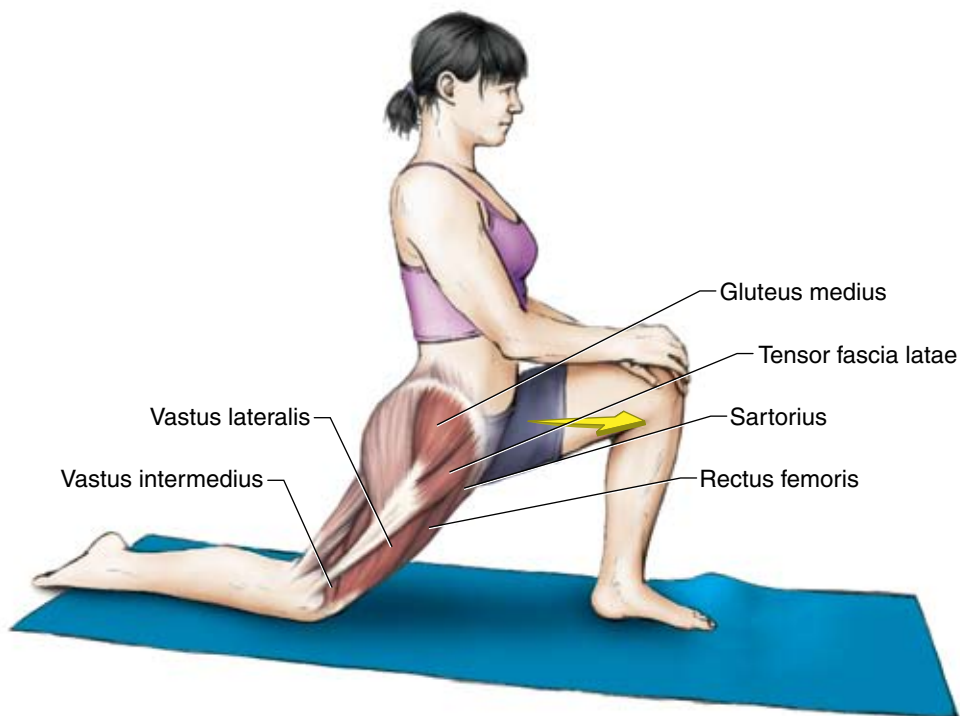
Most-stretched muscles: Gluteus maximus, semitendinosus, semimembranosus, gracilis, adductor magnus, adductor brevis, adductor longus, sartorius, erector spinae, latissimus dorsi.

Lesser-stretched muscles: Posterior deltoids, triceps brachii, lower trapezius, teres minor, teres major, infraspinatus.

Commentary

Keep the knees straight, bend the trunk forward from the hip joint, and keep the trunk as a straight unit (no back curve). By increasing the height of the table, bench, or other object 1 to 2 feet (30 to 61 cm) above the hips, you will have additional benefits to the noted muscle groups.

One-Leg Kneeling Knee Extensor Stretch



Technique

Step forward with the left leg and bend the knee at about a 90-degree angle. Keep the left knee positioned above the left ankle.

Extend the right leg behind the torso and touch the floor with the right knee; the lower leg lies on the floor.

Hold on to an object or place the hands on the left knee to maintain balance.

Move the hips forward, pushing the left knee in front of the left ankle and dorsiflexing that ankle.

Muscles Stretched

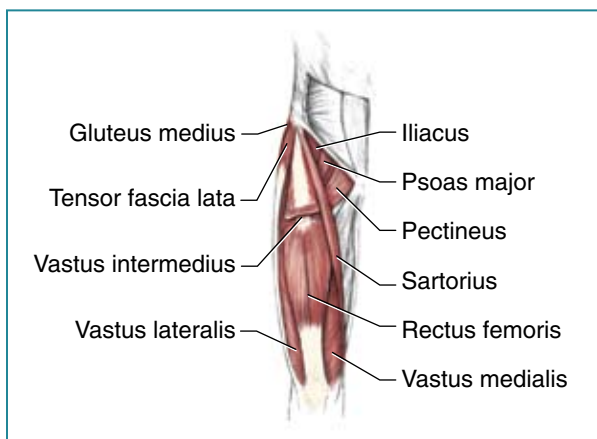
Most-stretched muscles: Right vastus medialis, right vastus intermedius, right vastus lateralis, middle and upper right sartorius, right rectus femoris, right psoas major, right iliacus, right tensor fascia lata.

Lesser-stretched muscles: Right pectineus, anterior part of right gluteus medius.

Commentary

Move slowly to the stretched position and keep the left knee pointing forward. Do not let the left knee point to either side or let the right knee move along the floor. While the hips are placed in the forward direction, arching the back can increase the stretch on the muscles.

One-Leg Standing Hip Flexor and Knee Extensor Stretch

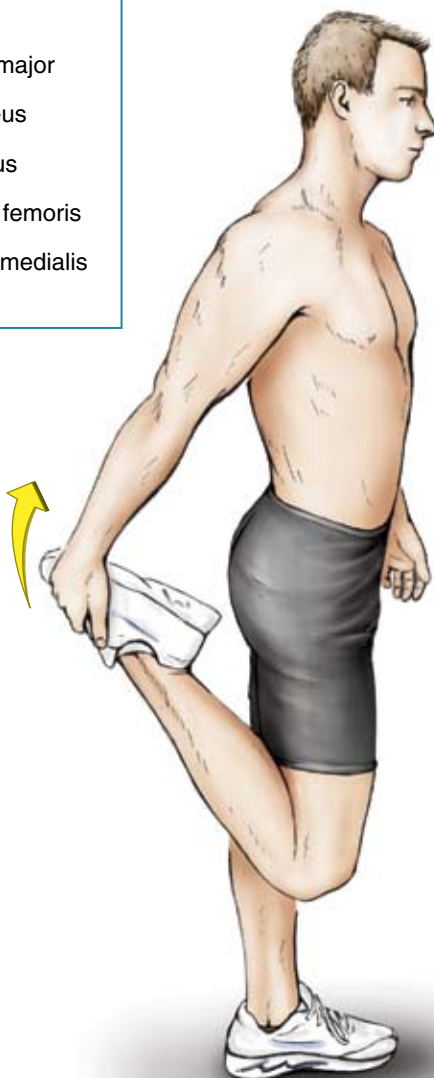


Technique

Stand upright with weight balanced on the left leg. Keep the left foot pointing straight forward and the knee almost straight. To help maintain balance, brace the left hand on a wall.

Bend the right knee; grasp the right foot or ankle tightly and pull the right heel backward and slightly upward to within 4 to 6 inches (10 to 15 cm) of the buttocks.

At the same time, push the hips forward.



Muscles Stretched

Most-stretched muscles: Right vastus medialis, right vastus intermedius, right vastus lateralis, middle and upper right sartorius, right rectus femoris, right psoas major, right iliacus, right tensor fascia lata.

Lesser-stretched muscles: Right pectineus, anterior part of right gluteus medius.

Commentary

When doing this stretch, be extra careful not to strain the knee structure by over-flexing the knee. Pull the ankle slowly in a more backward than upward direction, making sure that the hips also move forward. In other words, concentrate more on doing hip extension than on doing knee flexion. To place most of the stretch emphasis on the medial muscles (vastus medialis and pectineus), rotate the upper body away from the medial muscles (rotate the right side clockwise) when bending backward. To place most of the stretch emphasis on the lateral muscles (vastus lateralis and tensor fascia latae), rotate the upper body away from the lateral muscles (rotate the right side counterclockwise) when bending backward.

VARIATION

Supported One-Leg Standing Hip Flexor and Knee Extensor Stretch

You can also do this stretch by bracing the right foot on a table or beam. Because of the increased possibility of hyperflexing the knee, this is a more advanced stretch; you should do this stretch only if you have very flexible muscles.

Technique

- Stand with the back toward a padded table, bed, or soft platform that is below the height of the hips.
- Balance the weight on the left leg and bend the knee slightly.
- Bend the right knee and prop the right ankle on the rear support surface.
- Place both hands on the rear support surface 6 to 12 inches (15 to 30 cm) behind the buttocks.
- Move the torso backward slowly so that the heel of the right foot touches the buttocks. Make sure that the ankle and knee are comfortable.
- Push the hips forward and simultaneously arch the back by bending the shoulders toward the buttocks.

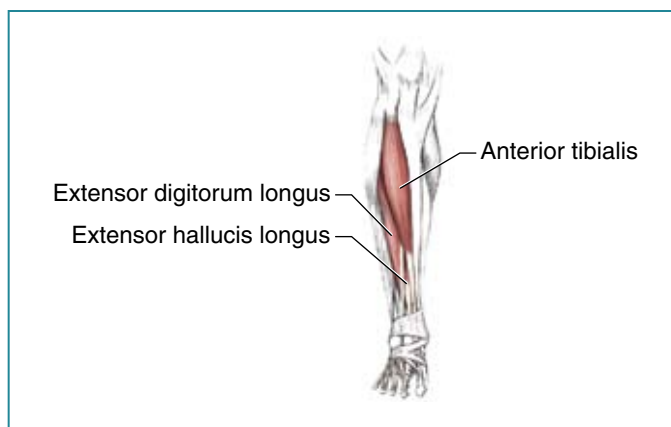
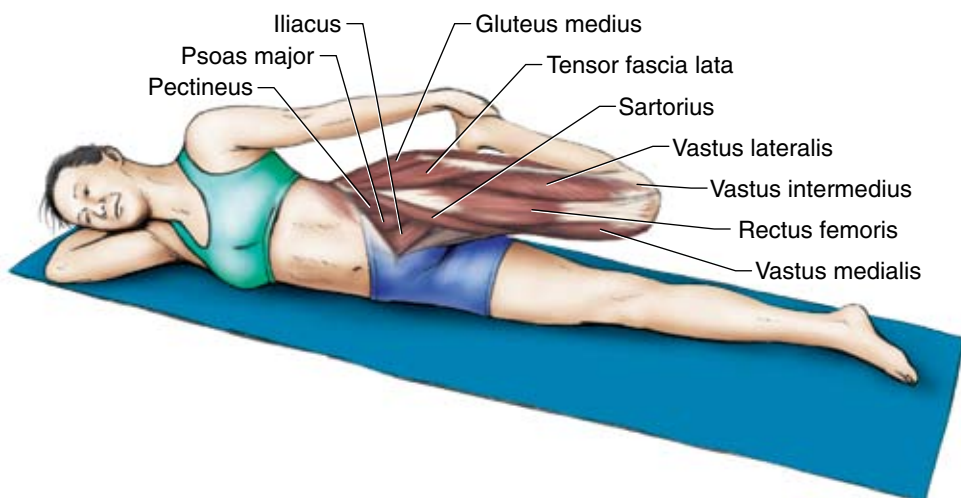


Muscles Stretched

Most-stretched muscles: Right vastus medialis, right vastus intermedius, right vastus lateralis, middle and upper right sartorius, right rectus femoris, right psoas major, right iliacus, right tensor fascia lata.

Lesser-stretched muscles: Right pectineus, anterior part of right gluteus medius.

Lying Hip Flexor and Knee Extensor Stretch



Technique

Lie on the right side of the body.

Bend the left knee and bring the left heel to within 4 to 6 inches (10 to 15 cm) of the buttocks.

Grasp the left ankle tightly and pull the leg backward close to your buttocks. However, do not bring the heel of the left ankle all the way to the buttocks.

Push the hip forward at the same time.

Muscles Stretched

Most-stretched muscles: Left vastus intermedius, left rectus femoris, left psoas major, middle and upper left sartorius.

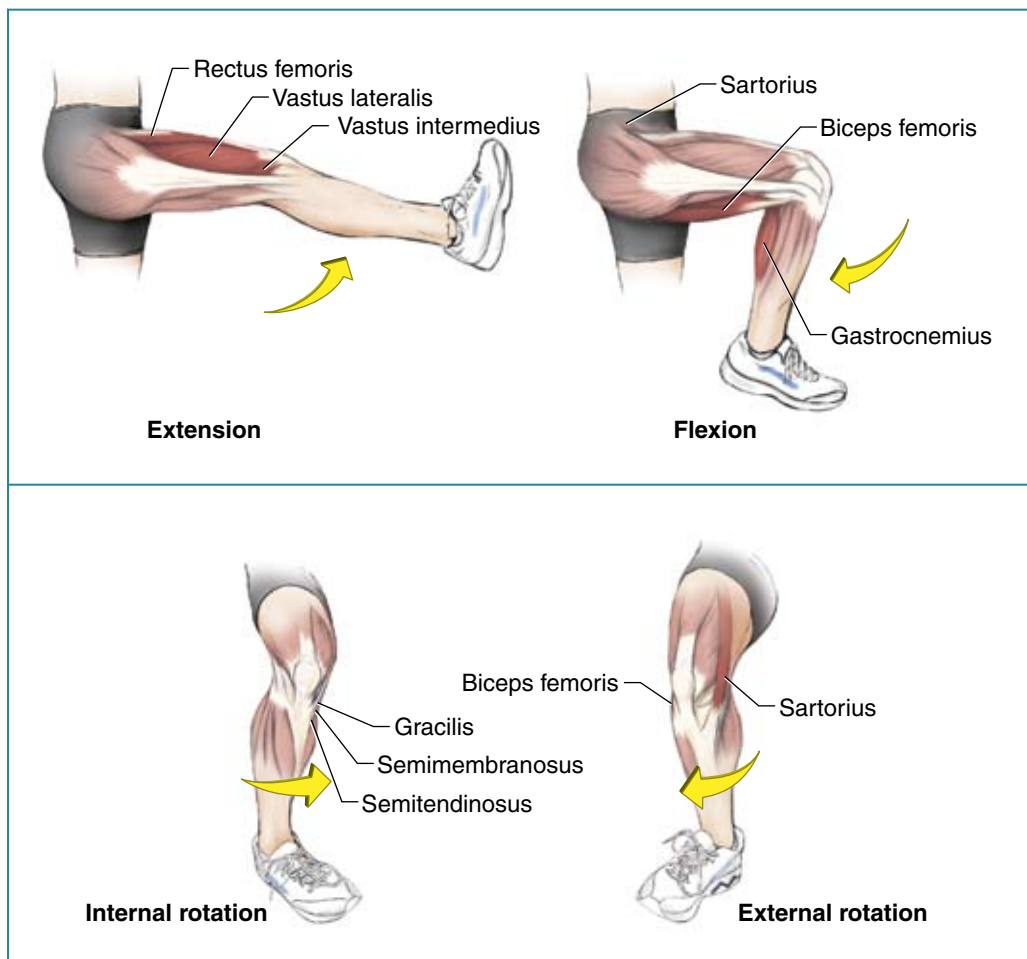
Lesser-stretched muscles: Left vastus medialis, left vastus lateralis, left tensor fascia lata, left pectineus, left iliacus, anterior part of left gluteus medius, left anterior tibialis, left extensor digitorum longus, left extensor hallucis longus.

Commentary

When doing this stretch, be extra careful not to strain the knee structure by over-flexing the knee. Pull the ankle slowly in a more backward direction than upward direction, making sure that the hips are also moved forward. In other words, concentrate more on doing hip extension than on doing knee flexion.

Knee and Thigh Muscle Movements

The stretches in this chapter are excellent overall stretches; however, not all of these stretches may be completely suited to each person's needs. The muscles involved in the various thigh and knee movements appear in the following table. To stretch specific muscles, the stretch must involve one or more movements in the opposite direction of the desired muscle's movements. For example, if you want to stretch the left biceps femoris, you would perform a movement that involves extension and internal rotation of the left leg. When a muscle has a high level of stiffness, you should use fewer simultaneous opposite movements (for example, to stretch a very tight biceps femoris, you could start by doing only knee extension). As a muscle becomes loose, you can incorporate more simultaneous opposite movements.



Muscle	Knee extension	Knee flexion	Internal rotation	External rotation
Rectus femoris	✓			
Vastus medialis	✓			
Vastus lateralis	✓			
Vastus intermedius	✓			
Sartorius		✓		✓
Biceps femoris		✓		✓
Semitendinosus		✓	✓	
Semimembranosus		✓	✓	
Gracilis		✓	✓	
Popliteus		✓	✓	
Gastrocnemius		✓		
Plantaris		✓		